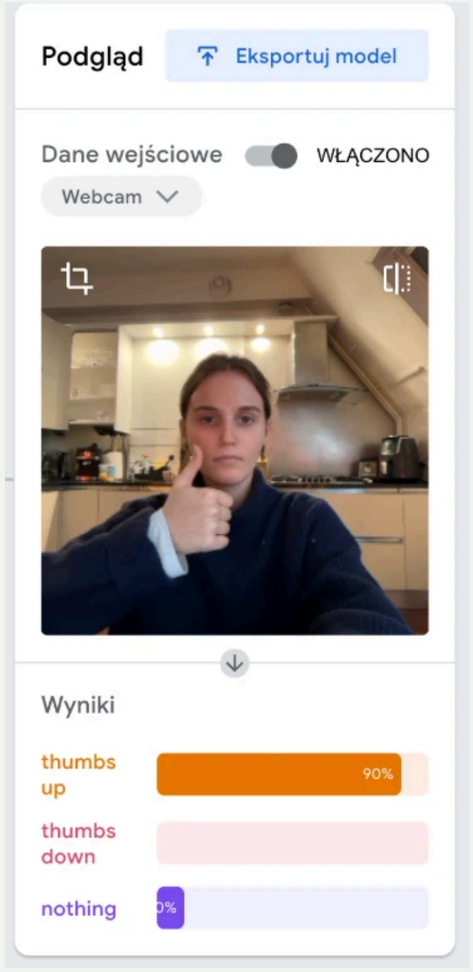
Computer Vision — Lecture 7 Retake

AI for Good · Nina Borutyńska · 25044923

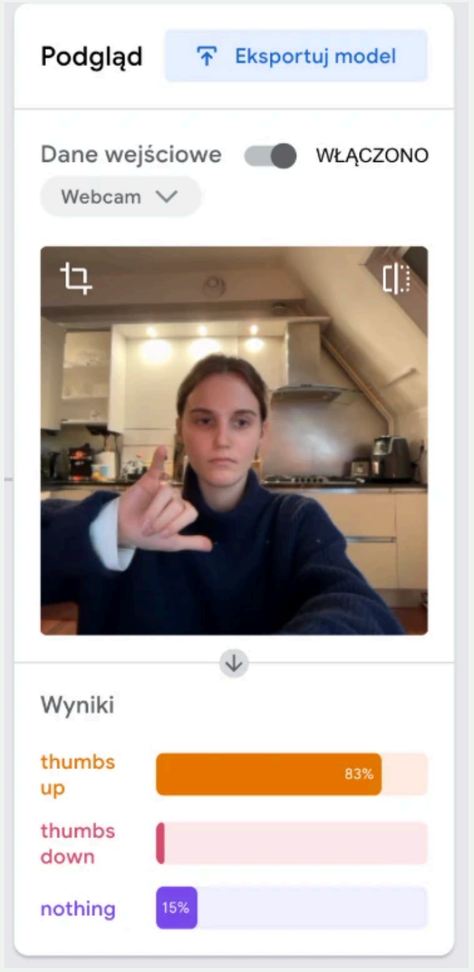


My Teachable Machine model

- Classes: thumbs up (52) · thumbs down (44) · nothing (45)
- Data: webcam photos, same kitchen, same light, only me
- Test: thumbs up 90% · thumbs down 100%
- Failure: another hand gesture → thumbs up 83%
- Likely cause: only 3 classes, it must pick one; it learned “a hand near my face”



✓ works · 90%



✗ fails · 83%

Example 1: MONOcam (Dutch police)

- Detects drivers holding a phone
- Benefit: safer roads
- Harm: every driver filmed, wrong flags
- Safeguard: two officers check every photo, non-violations deleted

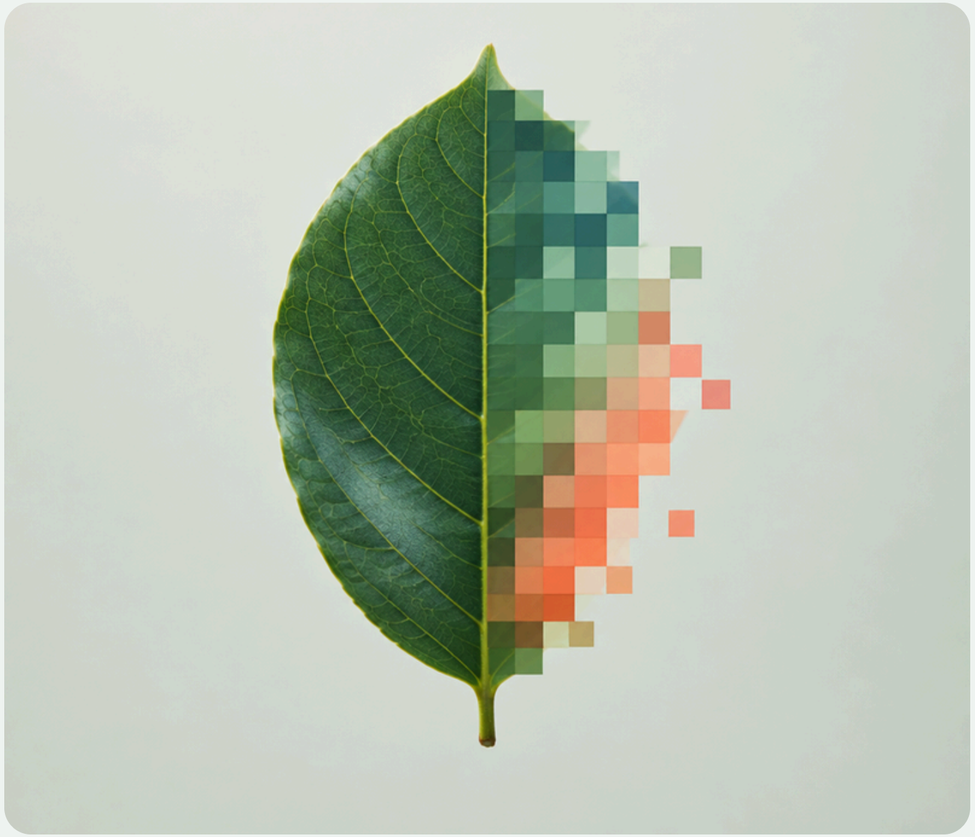


Example 2: Clearview AI



Scraped photos → face search for police · Harm: no consent, wrong matches · 2024: Dutch DPA fine €30.5 million

Images are numbers



23	41	66
84	128	172
117	203	241

Grayscale: 1 number per pixel (0 = black, 255 = white) · Colour: 3 numbers (R, G, B) · Video: images over time

Three preprocessing methods



Resize
faster · small details lost

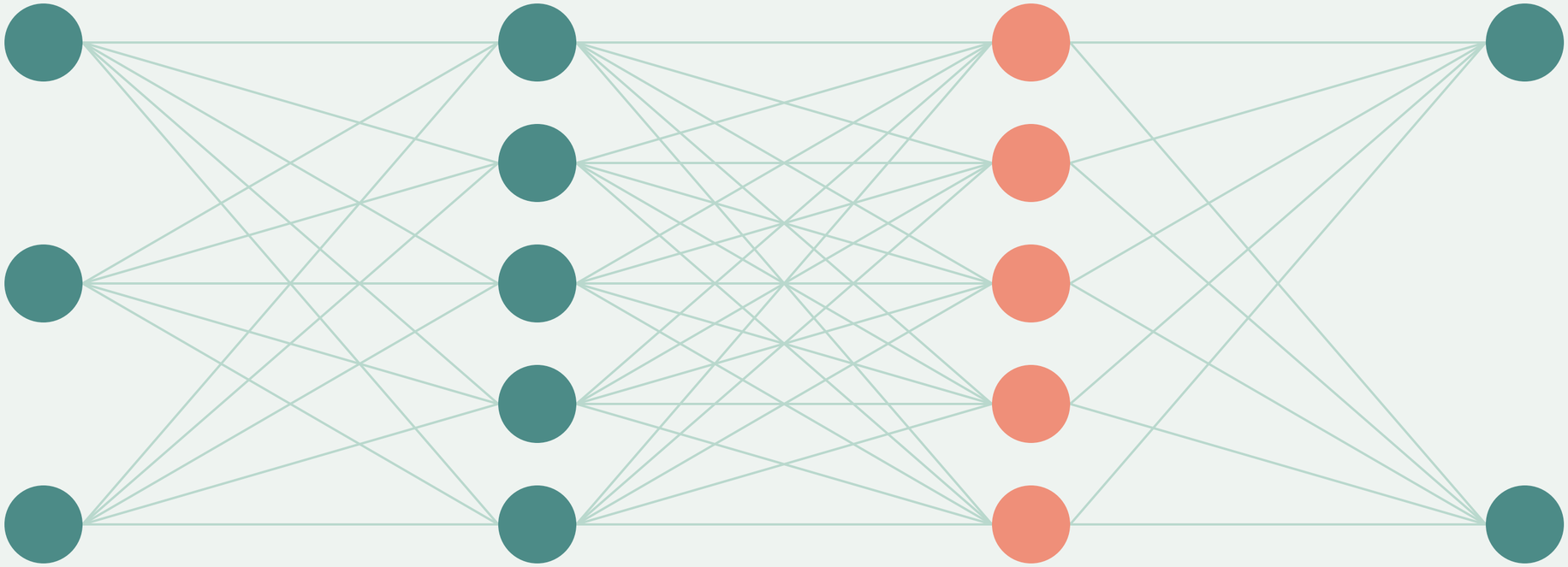


Grayscale
3x less data · colour lost



Contrast
clearer · more noise

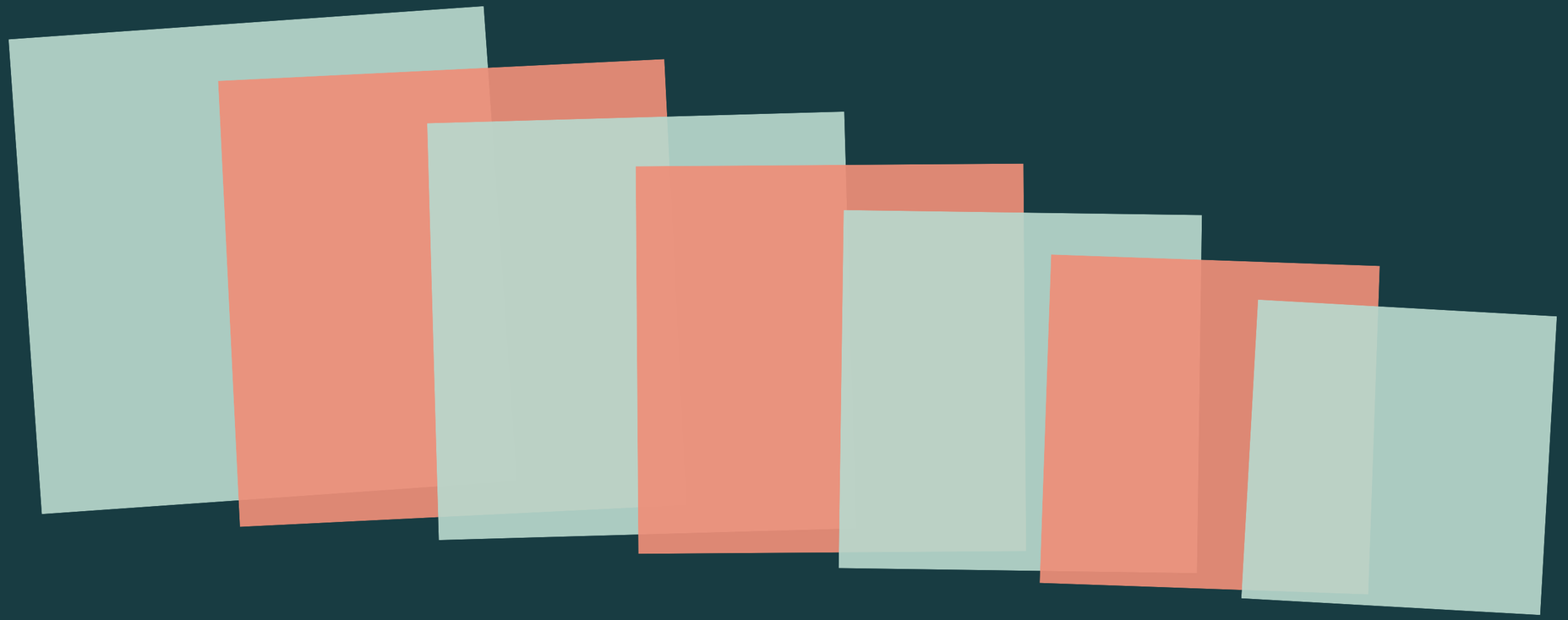
How a neural network learns



Predict → compare with the right answer → adjust weights → repeat

Images are hard: 224 × 224 × 3 ≈ 150,000 numbers · objects move · neighbours get lost

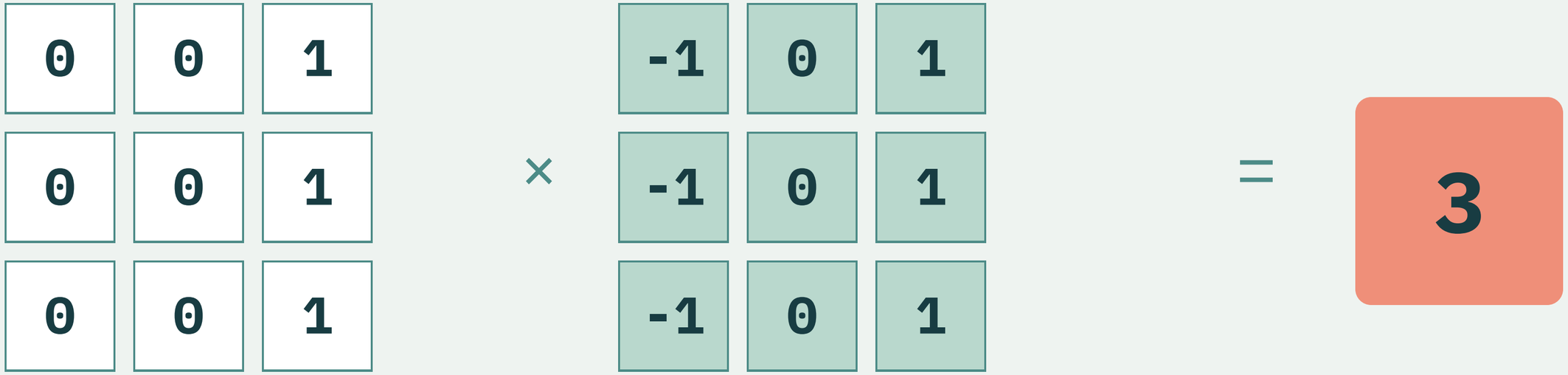
CNNs



Small filters slide over the image · same filter reused everywhere = fewer weights

Edges → shapes → objects

Convolution step by step



Multiply each pixel by the filter number and add up → 3 = edge found

Flat area → 0 = no edge · slide the filter → feature map

CNNs beyond images: BirdNET

Bird sound → spectrogram (time × frequency) → CNN → species

Cornell University · used to monitor biodiversity



Thank you

Sources: Lecture 7 slides (THUAS) · algoritmes.overheid.nl – MONOcam · Dutch DPA fine Clearview AI, Sept 2024 (NL Times, Jurist) · birdnet.cornell.edu